1. Aerenchyma:
Parenchyma which contains air in its intercellular spaces. It helps in aeration and buoyancy. Example: Nymphae and Hydrilla.

5. Prosenchyma:
Parenchyma cells became elongated, pointed and slightly thick walled. It provides mechanical support.

Parenchyma

2. Storage Parenchyma:
Parenchyma stores food materials. Example: Root and stem tubers.

4. Chlorenchyma
Parenchyma cells with chloroplast. Function is photosynthesis. Example: Mesophyll of leaves.

3. Stellate Parenchyma
Star shaped parenchyma. Example: Petioles of *Banana and Canna*.

Collenchyma (Gk. Colla-glue; enchyma – an infusion)

Collenchyma is a simple, living mechanical tissue. Collenchyma generally occurs in hypodermis of dicot stem. It is absent in the roots and also occurs in petioles and pedicels. The cells are elongated and appear polygonal in cross section. The cell wall is unevenly thickened. It contains more of hemicellulose and pectin besides cellulose. It provides mechanical support and elasticity to the growing parts of the plant. Collenchyma consists of narrow cells. It has only a few small chloroplast or none. Tannin maybe present in collenchyma.Based on pattern of pectinisation of the cell wall, there are three types of collenchyma.

Types of Collenchyma

1. Angular collenchyma
It is the most common type of collenchyma with irregular arrangement and thickening at the angles where cells meet. Example: Hypodermis of *Datura* and *Nicotiana*

2. Lacunar collenchyma
The collenchyma cells are irregularly arranged. Cell wall is thickened on the walls bordering intercellular spaces. Example: Hypodermis of *Ipomoea*

3. Lamellar collenchyma
The collenchyma cells are arranged compactly in layers(rows). The Cell wall is thickened only at tangential wall. devoid of thickened at radial walls. These thickening appear as successsive tangential layers. Example: Hypodermis of Helianthus

Diagramatic structures

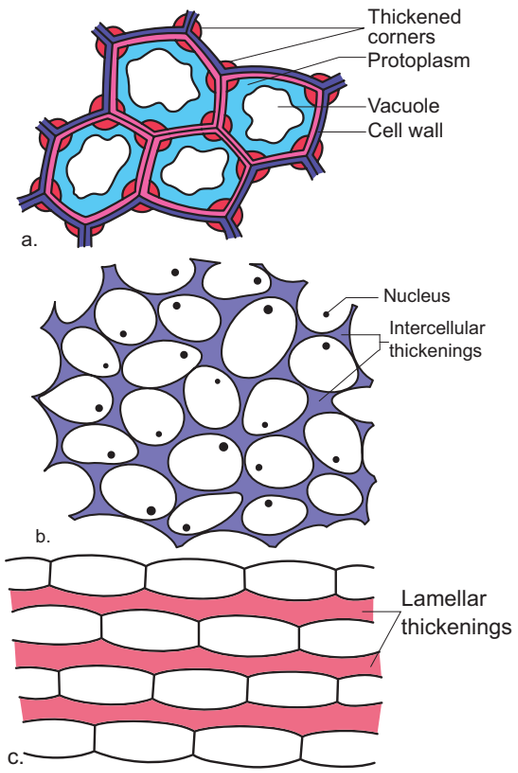


Figure 9.6: Types of Collenchyma
a) Angular collenchyma, b) Lacunar collenchyma, c) Lamellar collenchyma

11th Botany_EM_Chapter 9.indd 198 06/01/2023 20:02:02